

**Exercise 2.16.** Assume that  $a \equiv b \pmod{n}$  and  $c \equiv d \pmod{n}$ . Prove the following.

(a)  $a - c \equiv b - d \pmod{n}$ .

(b)  $a \cdot c \equiv b \cdot d \pmod{n}$ .

Solution.

We are given

$$a \equiv b \pmod{n}$$

and

$$c \equiv d \pmod{n}.$$

By the definition of congruence,

$$n \mid (a - b)$$

and

$$n \mid (c - d).$$

We will prove both statements from this definition.

(a) Prove  $a - c \equiv b - d \pmod{n}$

Since

$$n \mid (a - b)$$

and

$$n \mid (c - d),$$

there exist integers  $k$  and  $\ell$  such that

$$a - b = kn$$

and

$$c - d = \ell n.$$

Subtract the second equation from the first:

$$(a - b) - (c - d) = kn - \ell n.$$

Simplify:

$$a - b - c + d = (k - \ell)n.$$

Rearrange:

$$(a - c) - (b - d) = (k - \ell)n.$$

Therefore,

$$n \mid [(a - c) - (b - d)].$$

By the definition of congruence,

$$\boxed{a - c \equiv b - d \pmod{n}}.$$

**(b) Prove  $a \cdot c \equiv b \cdot d \pmod{n}$**

Again, we have

$$a - b = kn$$

and

$$c - d = \ell n.$$

We want to show

$$n \mid (ac - bd).$$

Start by factoring  $ac - bd$ :

$$ac - bd = ac - bc + bc - bd.$$

Group the terms:

$$ac - bd = c(a - b) + b(c - d).$$

Substitute

$$a - b = kn$$

and

$$c - d = \ell n :$$

$$ac - bd = c(kn) + b(\ell n).$$

Factor out  $n$ :

$$ac - bd = n(ck + b\ell).$$

Since  $ck + b\ell$  is an integer,

$$n \mid (ac - bd).$$

Therefore,

$$\boxed{ac \equiv bd \pmod{n}}.$$

**The key idea**

The definition

$$a \equiv b \pmod{n} \iff n \mid (a - b)$$

lets us manipulate congruences by manipulating the differences. Thus, congruence modulo  $n$  is preserved under **subtraction** and **multiplication**.